Search strategy

Database	Publisher	Date	Results
Medline	Ovid	21-10-2020	498
Embase	Elsevier	22-10-2020	1338
CENTRAL, Cochrane	Wiley	22-10-2020	313
Total with duplicates			2149
Total without duplicates			1554

Ovid MEDLINE(R) ALL / PubMed(R) <1946 to Present>

1 exp Gastroesophageal Reflux/ (26735)

2 exp Heartburn/ (2076)

3 ((Gastroesophageal or gastro-esophageal or gastro-oesophageal) adj5 reflux).tw. (24116)

4 (acid adj4 reflux).tw. (3471)

5 GERD.tw. (8566)

6 GORD.tw. (824)

7 heartburn.tw. (5071)

8 pyros#s.tw. (277)

9 reflux.tw. (53805)

10 regurgitation.tw. (39109)

11 or/1-10 (99313)

12 exp Endoscopic Mucosal Resection/ (1730)

13 ((antireflux or anti-reflux) adj9 ablation).tw. (31)

14 ARMA.tw. (707)

15 mucosectomy.tw. (801)

16 ARMS.tw. (61259)

17 ARAT.tw. (454)

18 "mucosal resection".tw. (2958)

19 "submucosal dissection".tw. (4578)

20 ESD.tw. (4480)

21 EMR.tw. (6641)

22 argon.tw. (15906)

23 APC.tw. (23271)

24 or/12-23 (115455)

25 11 and 24 (508)

26 exp Animals/ (23524624)

27 animal*.tw. (1100039)

28 26 or 27 (23657195)

29 exp Humans/ (18778506)

30 human*.tw. (2767083)

31 29 or 30 (19385896)

32 28 not 31 (4592890)

33 25 not 32 (498)

Embase Elsevier

#27 #11 AND #24 AND [humans]/lim AND [embase]/lim 1,338

#26 #11 AND #24 AND [humans]/lim1,417

#25#11 AND #24 1,569

#24 #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 195,636

#23 apc:ab,ti 34,865

#22 argon:ab,ti 17,206

#21 emr:ab,ti 16,405

#20 esd:ab,ti 10,086

#19 'submucosal dissection':ab,ti 9,161

#18 'mucosal resection':ab,ti 6,181

#17 ara:ab,ti 13,776

#16 arms:ab,ti 99,459

#15 mucosectomy:ab,ti 1,268

#14 arma:ab,ti 1,084

#13 ((antireflux OR 'anti reflux') NEAR/9 ablation):ab,ti 40

#12 'endoscopic mucosal resection'/de 6,720

#11 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 170,927

#10 regurgitation:ab,ti 64,026

#9 reflux:ab,ti 81,220

#8 pyros?s:ab,ti 468

#7 heartburn:ab,ti 8,952

#6 gord:ab,ti 1,470

#5 gerd:ab,ti 17,743

#4 (acid NEAR/4 reflux):ab,ti 5,826

#3((gastroesophageal OR 'gastro esophageal' OR 'gastro oesophageal') NEAR/5 reflux):ab,ti 36,279

#2 'heartburn'/de 14,594

#1'gastroesophageal reflux'/de 57,463

CENTRAL, Cochrane (Wiley)

#1	MeSH descriptor: [Gastroesophageal Reflux] explode all trees	1929
#2	MeSH descriptor: [Heartburn] explode all trees	389
#3	(((Gastroesophageal or gastro-esophageal or gastro-oesophageal) NEAR/5 reflux)):ti,ab,kw	4517
#4	(acid NEAR/4 reflux):ti,ab,kw	608
#5	(GERD):ti,ab,kw	1725
#6	(GORD):ti,ab,kw	190
#7	(heartburn):ti,ab,kw	2293
#8	(pyros*s):ti,ab,kw	82
#9	(reflux):ti,ab,kw	7396
#10	(regurgitation):ti,ab,kw	2521
#11	{OR #1-#10}	10447
#12	MeSH descriptor: [Endoscopic Mucosal Resection] explode all trees	78
#13	(((antireflux or anti-reflux) NEAR/9 ablation)):ti,ab,kw	8
#14	(ARMA):ti,ab,kw	133
#15	(mucosectomy):ti,ab,kw	66
#16	(ARMS):ti,ab,kw	49256
#17	(ARA):ti,ab,kw	1462
#18	("mucosal resection"):ti,ab,kw	419
#19	("submucosal dissection"):ti,ab,kw	784
#20	(ESD):ti,ab,kw	887
#21	(EMR):ti,ab,kw	939

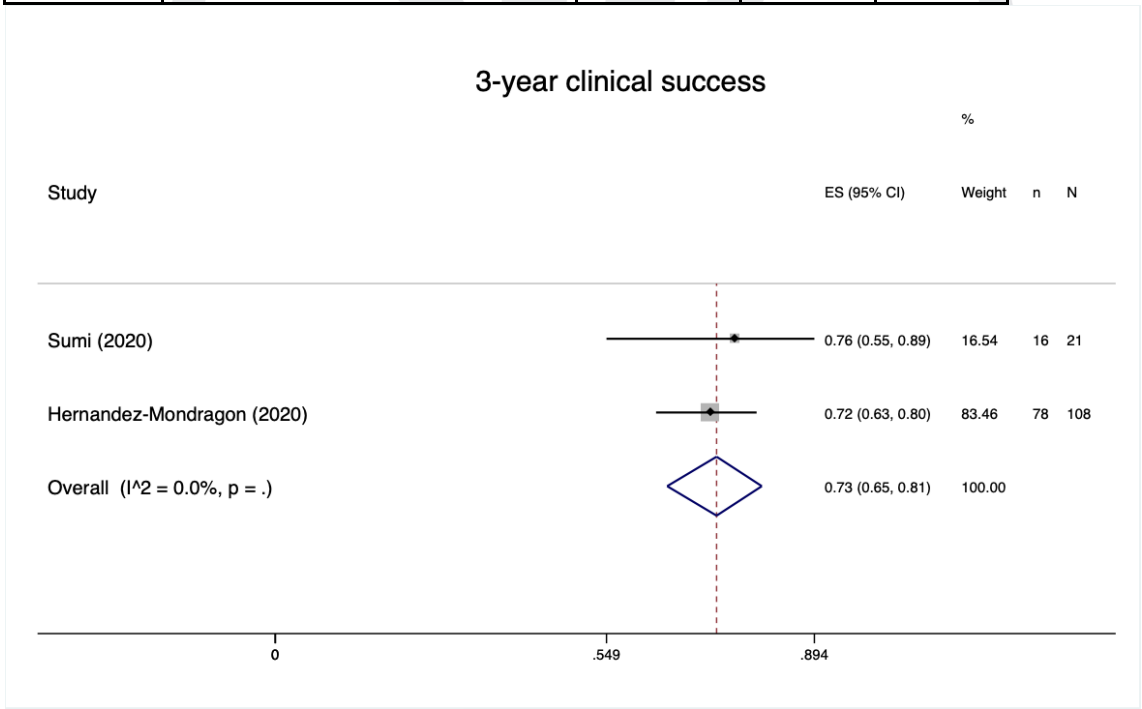
#22	(argon):ti,ab,kw	975
#23	(APC):ti,ab,kw	847
#24	{OR #12-#23}	54153
#25	#11 AND #24	319
#26	#25 in Trials	313

We handsearched abstracts from annual meetings (2013 to November 2020) of the following scientific societies:

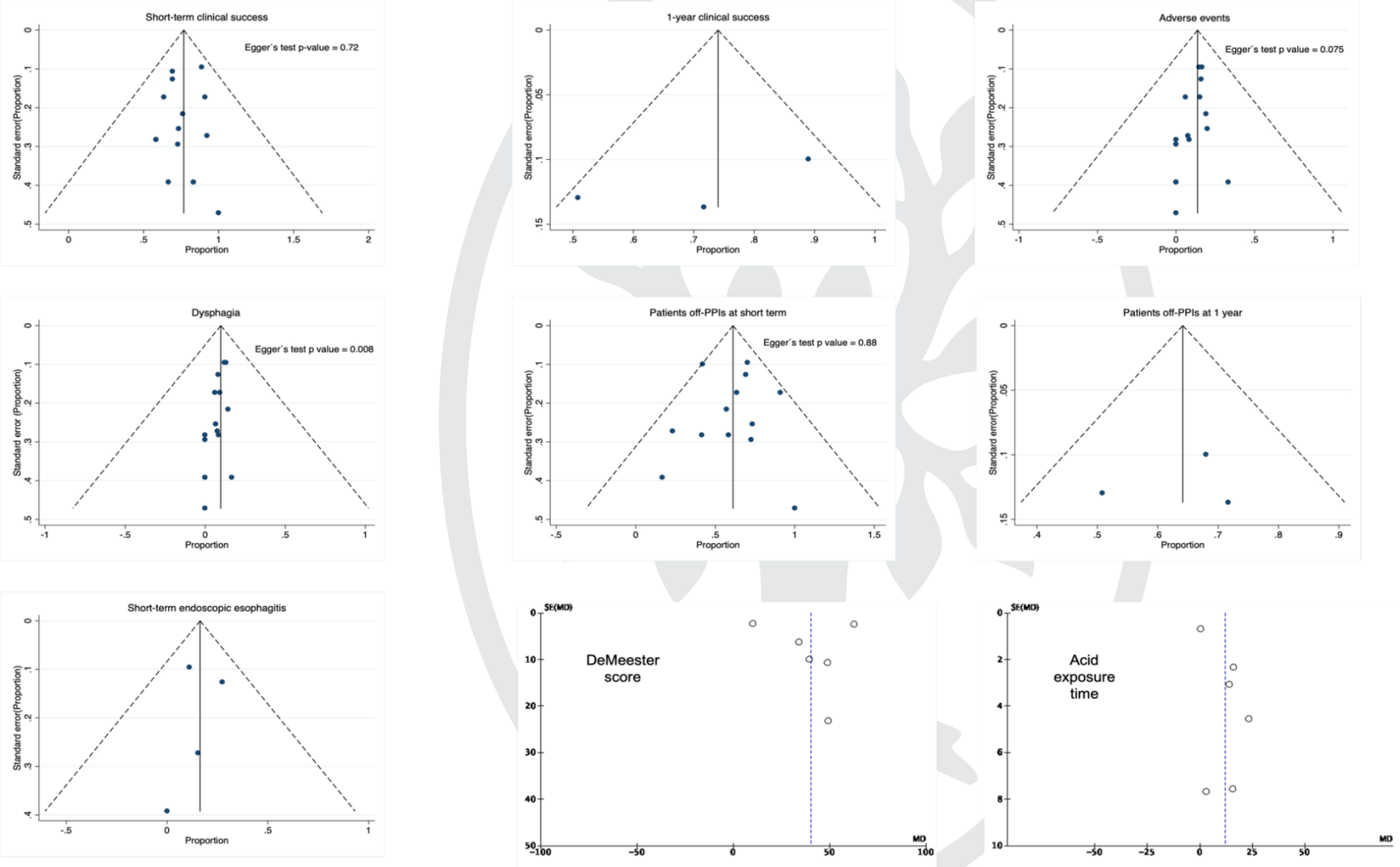
- European Society of Gastrointestinal Endoscopy (ESGE).
- American Society of Gastrointestinal Endoscopy (ASGE).
- United European Journal (UEG).
- Asian Pacific Society of Digestive Endoscopy (A-PSDE).

3-year outcomes

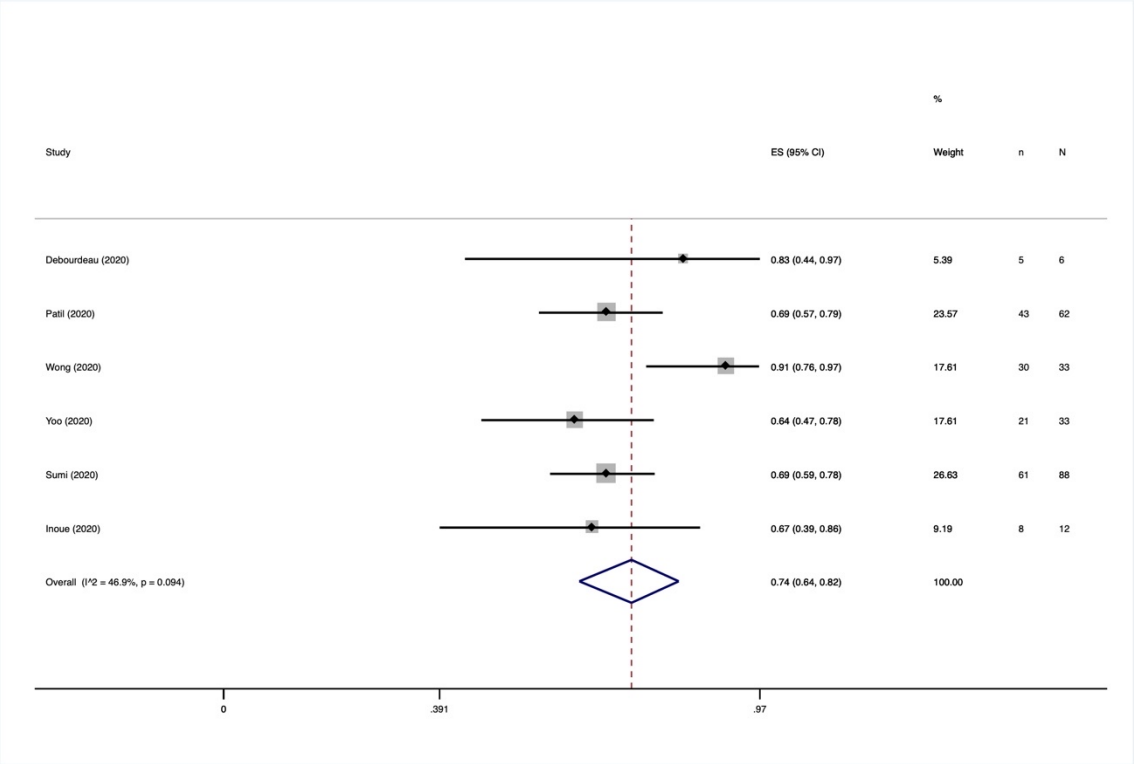
Outcome	3 years			
	No. of studies; N	Pooled proportion	95% CI (%)	I^2 (%)
Clinical success	2; 129	73%	75–81	0
GERD-HRQL	No data available			
GerdQ	No data available			
Patients without PPIs	1;84	66 out of 84 (78.5%)	Not estimated	Not estimated



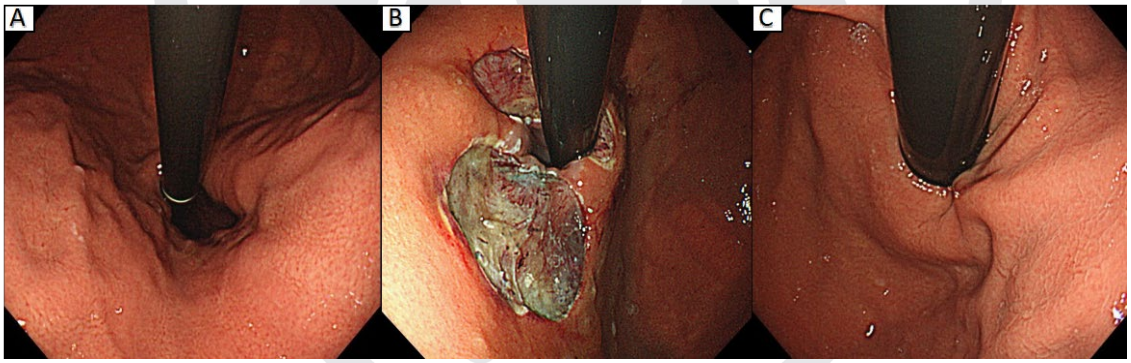
Supplementary Fig. 1 Publication bias.



Supplementary Fig. 2 Forest plot of clinical success defined as a decrease of 50% or more in a validated clinical score.



Supplementary Fig. 3 Antirelux mucosectomy (ARMS).

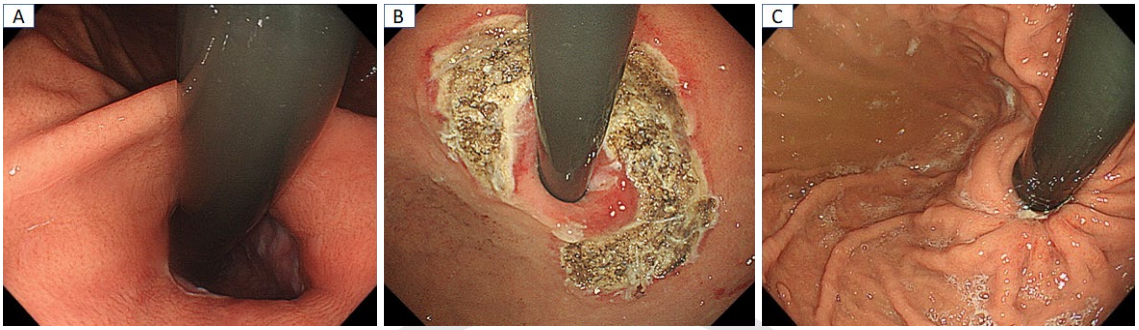


a. Endoscopic view in retroflexion before antireflux mucosectomy (ARMS).

b. Immediately post-ARMS. Endoscopic view of Butterfly resection with cap-assisted endoscopic mucosal resection. One cm of healthy mucosa is spared in the lesser and greater curvature of the stomach.

c. 2 months after ARMS.

Supplementary Fig. 4 Antireflux mucosal ablation (ARMA)



a. Endoscopy in retroflexion before antireflux mucosal ablation (ARMA).

b. Immediately post-ARMA. 320º ablation by argon plasma coagulation.

c. 1 month after ARMA. Ulcer scar produces a fundoplication-like morphology of the gastroesophageal valve.

Supplementary Table 1 Additional baseline characteristics of the included studies.

First author, year	Age (years)	Sex (M:F)	Technical success definition	Sedation protocol	Procedure duration (minutes)	Inpatient or outpatient procedure	Hospitalization duration	Technical success
Ota, 2014	51.6 (14.9)	11:2	N/S	N/S	67.7	N/S	N/S	N/S
Bapaye, 2017	40.8	11:4	N/S	N/S	N/S	N/S	N/S	N/S
Shah, 2017	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S
Vasilevskiy, 2017	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S
Ortega, 2019	41	N/S	N/S	N/S	45	N/S	1.5	N/S
Mohan, 2019	N/S	N/S	N/S	N/S	N/S	N/S	< 2 days	N/S
Inoue, 2020	Median 54.5 (range 29-75)	7:5	N/S	Propofol sedation or general anesthesia (ETI)	Median 40.3 (range 20-69)	Inpatient and outpatient	4 days (0-7)	100%
Monino, 2020	56.8 (14.4)	11:13	N/S	General anesthesia (ETI)	35 (11)	Outpatient	0 days	100%
Yoo, 2020	51.3 (16.3)	22:11	N/S	Propofol and midazolam sedation	31.2 median (range 50-80)	N/S	N/S	N/S
Hernández-Mondragón, 2020	55.6 (13.4)	3:3	N/S	Sedation	27.4 (6.5)	Outpatient	N/S	N/S
Sumi, 2020	54 (15.7)	63:45	N/S	Propofol sedation or	54.7 (27)	Inpatient and outpatient	N/S	100%

				general anaesthesia (ETI)				
Patil, 2020	36 (9.9)	44:18	Succesful completion of resecting gastric cardia with adequate hemostasis	General anaesthesia (ETI)	N/S	Inpatient	3 days (2-5)	100%
Debourdeau, 2020	44 (8.2)	1:5	Completed mucosectomy visible in direct vision and retroflexión	General anaesthesia (ETI)	All < 40	Outpatient	N/S	100%
Wong, 2020	55 (17)	11:22	N/S	General anaesthesia (ETI)	36 (13)	Outpatient	2.9 hours (1.1)	100%
Hernández-Mondragón, 2020	Median 36.5 (IQR: 18-78)	61:47	N/S	Propofol and fentanyl sedation	Median 35.5 (IQR: 22-51)	Outpatient	N/S	100%

M, male; F, female; SD, standard deviation; IQR, interquartile range; ETI, endotracheal intubation; N/S, not specified.

Numbers are mean (SD) unless otherwise specified; **numbers are median (range) unless otherwise specified.

Supplementary Table 2 Data on secondary outcomes.

First author, year	Questionnaires		PPs dose modification after procedure		Acid exposure time		DeMeester score		Hill grade		Erosive esophagitis (Los Angeles classification)	
	Baseline	Post	Dose reduction	Off PPIs	Baseline	Post	Baseline	Post	Baseline	Post	Baseline	Post
Ota, 2014*	N/S	N/S	**6 mo:** 3/13	**6 mo:** 3/13	10.4 (15.5)	**6 mo:** 7.4 (10.6)	N/S	N/S	N/S	N/S	B:2 C/D: 2/1	**6 mo:** any grade 2/13
Bapaye, 2017*	GERD-HRQL: 40.4	**4-6 wk**: 7.6	**4-6 wk**: 4/15	**4-6 wk:** 11/15	N/S	N/S	85.8	**4-6 wk:** 5.9	2.8	**4-6 wk:** 1.6	N/S	N/S
Shah, 2017*	N/S	N/S	N/S	7/12	N/S	N/S	91.7	18.8	N/S	N/S	N/S	N/S
Vasilevskiy, 2017	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S
Ortega, 2019*	GerdQ: 19	3 mo: 3	N/S	**3 mo:** 4/4	N/S	N/S	37.1	N/S	N/S	N/S	N/S	N/S
Mohan, 2019	N/S	N/S	N/S	**6 mo:** 8/11	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S
Inoue, 2020**	GERD-HRQL: 30.5 (16-45)	**2 mo:** 12 (0-27)	N/S	**2 mo:** 7/12	9	**2 mo:** 0.5 5/8 with AET< 4.2%	60.8	**2 mo:** 11.5 5/8 with DMS< 14.7	3 (2-3)	**2 mo:** 1	A/B: 3/12	N/S
Monino, 2020**	GERD-HRQL: 23.5 GerdQ: 12.5	**6 mo:** 16.5 **6 mo:** 9	**6 mo:** 4/21	**6 mo:** 12/21	N/S	N/S	N/S	N/S	1 (1-3)	**6 mo:** 1 (1-3)	Any grade: 0/21	N/S

Yoo, 2020**	GerdQ: 11 (5-18)	**6 mo:** 6 (3-16)	**6 mo:** 10/33	**6 mo:** 21/33	2.7 (0-12.3)	**6 mo:** 0.9 (0-10)	19 (5-46.1)	**6 mo:** 4 (0.8-40)	3 (1-4)	**6 mo:** 1 (1-3)	N/S	N/S
Hernández-Mondragón, 2020*	N/S	N/S	N/S	N/S	19.8 (5.4)	**3 mo:** 3.7 (1.9)	44.3 (14.6)	**3 mo:** 10.2 (5.5)	Median 2.5 (range 1-3)	**3 mo:** median 1 (range 1-1)	A/B: 3/6 C/D: 1/6	**3 mo:** any grade 0/6
Sumi, 2020*	GerdQ: 9.4 (2.7)	**2-6 mo:** 6.6 (2.5)	N/S	**6 mo:** 42/100 **12 mo:** 30/59	20.8 (24.3)	**2-6 mo:** 6.9 (10.4) 14/27 with AET< 4.2%	64.4 (75.7)	**2-6 mo:** 24.9 (36)	Median 3 (range 1-4) 13/27 with DMS< 14.7	N/S	N/S	N/S
Patil, 2020*	GerdQ: 10.6 (1.9)	**2 mo:** 3.4 (1.5)	**2 mo:** 12/62 **6 mo:** 9/56 **12 mo:** 8/53	**2 mo:** 43/62 **6 mo:** 40/56 **12 mo:** 38/53	N/S	N/S	76.8 (18.3)	**2 mo:** 14.3 (6.1) 45/62 with DMS < 14.7	Median 3	**2 mo:** median 1 (range 1-2)	A/B: 57/52 C/D: 5/62	**2 mo:** A/B 17/62; C/D 0/62
Debourdeau, 2020*	GERD-HRQL: 30.6 GerdQ: 13.3 (1.2)	**3 mo:** 6.8 **3 mo:** 6.2 (4.4)	**3 mo:** 2/6	**3 mo:** 1/6	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S
Wong, 2020*	GERD-HRQL: 16 (12)	**6 mo:** 6 (7.1) **12 mo:** 2.1 (2) **24 mo:** 6 (3.1)	N/S	30/33	N/S	N/S	Median 20 (IQR: 13-26)	N/S	Median 3 (IQR: 2-3)	Median 1 (IQR: 1-3)	N/S	N/S

Hernández-Mondragón, 2020**	GERD-HRQL: 36.5 (IQR: 20-74)	**3 mo:** 10 (IQR: 3-38) **6 mo:** 7 (IQR: 1-25) **12 mo:** 8 (IQR 3-40) **24 mo:** 9 (IQR: 0-39) **36 mo:** 10.5 (IQR: 3-38)	N/S	**3 mo:** 76/108 **6 mo:** 74/102 **12 mo:** 68/100 **24 mo:** 72/93 **36 mo:** 66/84	Median 18.8 (IQR: 6.3-66.4)	**3 mo:** 2.8 (IQR: 0.2-13.5) 96/108 with AET< 4.2% **6 mo:** 2.9 (IQR: 0.4-13.5) **12 mo:** 2.6 (IQR: 0.3-7.5) **24 mo:** 2.5 (IQR: 0.3-7.3) **36 mo:** 2.2 (IQR: 0.2-6.5)	42.5 (IQR: 15.4-154)	**3 mo:** 9.1 (IQR: 1-55.4) **6 mo:** 7.3 (IQR: 17-33.1) **12 mo:** 7.8 (IQR: 1.6-18.4) **24 mo:** 9.8 (1.3-22.4) **36 mo:** 9.4 (IQR: 1.2-89.8)	3 (range 2-3)	**3 mo:** 1 (1-2) **6 mo:** 1 (1-2) **12 mo:** 1 (1-1) **24 mo:** 1 (1-1) **36 mo:** 1 (1-1)	A/B: 61/97 C/D: 38/97	**3m:** A/B 12/108; C/D 0/108 **6m:** A/B 11/102; C/D 1/102 **12m:** A/B 8/100; C/D 1/100 **24m:** A/B 5/93; C/D 1/93 **36m:** A/B 3/84; C/D 1/84

AET, acid exposure time; DMS, DeMeester score; GERD-HRQL, gastroesophageal reflux health-related quality of life questionnaire; PPIs, proton pump inhibitors; IQR, interquartile range; mo, months; wk, weeks.

Follow-up time is specified for each comparison. Continuous data are expressed as mean (standard deviation) unless otherwise specified.

Supplementary Table 3 Detailed description of adverse events.

First author, country, year	Sample size	Total adverse events	Dysphagia requiring dilation	Mean no. of dilations per patient	Dilator caliber (mm)	Clinically significant bleeding		Perforation		Aspiration pneumonia	Mediastinitis	Muscle injury without perforation
						Intraprocedural	Delayed	Intraprocedural	Delayed			
Ota, Japan, 2014	13	1	1	N/S	N/S	0	0	0	0	0	0	0
Bapaye, India, 2017	15	3	1	N/S	N/S	0	0	0	0	0	0	2
Shah, India, 2017	12	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S
Vasilevskiy, Russia, 2017	6	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S	N/S
Ortega, Spain, 2019	7	0	0*	0	0	0	0	0	0	0	0	0
Mohan, India, 2019	11	0	0	0	0	0	0	0	0	0	0	0
Inoue, Japan, 2020	12	2	1	1	13.5	0	0	0	0	0	0	1
Monino, France, 2020	24	7	3**	1.7	12	0	1	0	0	0	0	0
Yoo, South Korea, 2020	33	2	2	1	N/S	0	0	0	0	0	0	0
Hernandez-Mondragón, Mexico, 2020	10	1	0***	0	0	0	0	0	0	0	0	0
Sumi, Japan, 2020	109	16	13****@	0	0	0	2	1	0	0	0	0
Patil, India, 2020	62	10	5	1	12-15	0	0	1	1	3	0	0
Debourdeau, France, 2020	6	2	1	1	13.5	0	1	0	0	0	0	0
Wong, USA, 2020	33	5	3	2	N/S	0	1	1	0	0	0	0
Hernandez-Mondragón, Mexico, 2020	108	18	14	1.9	N/S	4	0	0	0	0	0	0

*4, ** 21, ***6, and ****88 patients with available follow-up to assess dysphagia, respectively. @Authors only included those patients who needed 3 or more dilations.

Supplementary Table 4 Sensitivity analyses according to esophageal mucosal involvement during ARMS and ARMA.

Outcome	Esophageal mucosa involvement				Gastric-cardiac only procedure				*P* value
	No. of studies; N	Pooled proportion	95% CI (%)	I^2 (%)	Nº of studies; N	Pooled proportion	95% CI (%)	I^2 (%)	
Short-term clinical success	5; 108	76%	66-84	0	9; 307	78%	66-88	69.1	0.74
1-year clinical success	1; 53	72%	58-82	0	2; 159	77%	70-83	0	0.45
Patients without PPIs in short term	4; 102	46%	21-71	78	8; 313	69%	53-83	81.9	0.16
Patients without PPIs at 1 year	1; 53	72%	58-82	0	2; 159	62%	54-69	0	0.20
Overall adverse events	5; 108	14%	7-22	0	9; 328	9%	5-14	13.6	0.30
Dysphagia requiring dilation	5; 108	7%	2-14	0	9; 328	8%	4-11	0	0.98

CI, confidence interval; PPIs, proton pump inhibitors.

Supplementary Table 5 Sensitivity analyses according to the type of publication.

Outcome	Full-text publication				Conference abstract				*P* value
	Nº of studies; N	Pooled proportion	95% CI (%)	I^2 (%)	Nº of studies; N	Pooled proportion	95% CI (%)	I^2 (%)	
Short-term clinical success	9; 376	78%	69-86	69.3	6; 54	74%	60-86	0	0.44
1-year clinical success	3; 209	72%	47-92	0	0	*	*	*	*
Patients without PPIs at short term	9; 388	59%	45-72	83.7	5; 42	73%	55-87	0.6	0.20
Patients without PPIs at 1 year	3; 212	64%	52-75	68	0	*	*	*	*
Overall adverse events	9; 397	14%	10-18	0	6; 54	2%	0-10	0	0.03
Dysphagia requiring dilation	9; 397	10%	7-13	0	6; 54	0%	0-6	0	0.04

CI, confidence interval; PPIs, proton pump inhibitors.

Supplementary Table 6 Sensitivity analyses according to study quality.

Outcome	Moderate-high quality				Poor quality				*P* value
	No. of studies; N	Pooled proportion	95% CI (%)	I^2 (%)	No. of studies; N	Pooled proportion	95% CI (%)	I^2 (%)	
Short-term clinical success	7; 303	74%	62-84	34.2	8; 127	82%	71-91	34.2	0.45
1-year clinical success	3; 209	72%	47-92	0	0	*	*	*	*
Patients without PPIs at short term	6; 309	62%	48 - 75	76.5	7; 121	63%	40-84	80.7	0.95
Patients without PPIs at 1 year	3; 212	64%	52 - 75	68	0	*	*	*	*
Overall adverse events	7; 324	13%	9-17	0	8; 127	8%	3-15	7	0.34
Dysphagia requiring dilation	7; 324	9%	5-12	0	8; 127	5	1-11	0	0.57

CI, confidence interval; PPIs, proton pump inhibitors.